

MA.912.GR.3 *Use coordinate geometry to solve problems or prove relationships.*

MA.912.GR.3.1

Benchmark

MA.912.GR.3.1 Determine the weighted average of two or more points on a line.

Benchmark Clarifications:

Clarification 1: Instruction includes using a number line and determining how changing the weights moves the weighted average of points on the number line.

Connecting Benchmarks/Horizontal Alignment

- MA.912.GR.1.1
- MA.912.GR.5.2

Terms from the K-12 Glossary

- Number Line

Vertical Alignment

Previous Benchmarks

- MA.7.AR.3

Next Benchmarks

- MA.912.NSO.3
- MA.912.FL.1.3
- MA.912.DP.6.7
- MA.912.LT.2.7
- MA.912.LT.3.3

Purpose and Instructional Strategies

In grade 7, student developed proportional reasoning skills. In Geometry, students learn the connection between weighted averages and proportional partitioning of line segments. In later course, weighted averages appear in a large number of contexts and they also are used in science courses.

- Instruction focuses on determining the weighted average of two points. This benchmark lays a foundation for weighted averages that will be extended into later courses through multiple pathways. For example, weighted averages can be used to calculate grades (i.e., final grade is 20% on midterm, 50% on final exam and 30% attendance); game

theory (i.e., mixed strategies); financial settings (i.e., portfolio consisting of 20% stock and 80% real estate); vectors (i.e., scalar multiplication closely related to partitioning line segments); and probability (i.e., expected value of a random variable is a weighted average of its possible values).

- Instruction includes the connection between weighted averages and partitioning line segments.
 - For example, students will learn that the question “Find the weighted average of the numbers -1 and 5 with weight $\frac{1}{4}$ on the first number and $\frac{3}{4}$ on the second number” is equivalent to the question “What point on the number line is $\frac{3}{4}$ the way from the point -1 to the point 5?”
 - In the prior example, the both questions could be solved by calculating $\frac{1}{4}(-1) + \frac{3}{4}(5)$ which equals 3.5 or by calculating $-1 + \frac{3}{4}(5 - -1)$ which equals 3.5. Students should be given the flexibility to use either method when solving problems.
- To assist students in their conceptual understanding, or visualization, of weighted averages, instruction includes the use of real weights on yard stick that is balanced on a pivot point. The purpose is not for students to compute the weighted average, but to visual how the weights affect the balance point. Students should explore how the change in weights changes the balance point. It is important to note that in real life, the weight of the yard stick will affect the balance point, if calculated.
 - For example, place three equal weights at 15 inches and 1 weight at 9 inches on the yard stick. If we can neglect the weight of the yard stick, then the balance point will be at 13.5 inches. Since the balance point of the yard stick is at 18 inches, the actual balance point in this experiment will move a little bit towards 18 inches from 13.5 inches, depending on how much the yard stick weighs compared to the weights.
 - For example, a teeter totter with an adult and a child will balance at the pivot point if the adult moves forwards/inward so that the weighted average of the two points is at the pivot point. If the adult weighs 200 pounds and the child weighs 50 pounds, then it is a 4: 1 partition. The weights for the weighted averages can be calculated as $\frac{200}{200+50}$, which equals $\frac{4}{5}$, and as $\frac{50}{200+50}$, which equals $\frac{1}{5}$.

Common Misconceptions or Errors

- Students may associate the larger weight to the longer segment when visualizing the problem.
- Students may multiply by the weights of the people, or things, instead of multiplying by the weights that lead to the weighted average, which must add up to 1. To help address this misconception, as in the teeter totter example above, students should realize that if they multiply by the weights of the people, then they would need to divide by the sum of the weights of the people.

Instructional Tasks

Instructional Task 1 (MTR.2.1, MTR.4.1, MTR.5.1)

Three numbers are provided below. Use these numbers to answer each question below.

0, 1, 2

Part A. What is the mean (m_1) of the three numbers?

Part B. Choose two of the numbers and determine their mean (m_2).

Part C. Determine the weighted average of m_2 and the third number using the weights $\frac{2}{3}$ and $\frac{1}{3}$. What do you notice?

Part D. Repeat Parts B and C with a different choice of the two numbers.

Part E. Repeat Parts A, B and C with any three real numbers, x , y and z . Share your answers with a partner. What do you notice?

Instructional Items

Instructional Item 1

What point on the number line is $\frac{7}{9}$ the way from the point -3.6 to the point 10?

**The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.*